PromethION

Ligation sequencing amplicons V14 (SQK-LSK114)

Oxford Nanopore Technologies

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Kit batch number .... **Flow cell number** .... **DNA Samples** ....

Checklist: End-prep

Materials	Consumables	Equipment
☐ 100-200 fmol amplicon DNA	☐ NEBNext® Ultra II End Prep Enzyme Mix from NEBNext® Ultra II End Repair Module (NEB, E7546)	☐ P1000 pipette and tips
☐ DNA Control Sample (DCS)	☐ NEBNext® Ultra II End Prep Reaction Buffer from NEBNext® Ultra II End Repair Module (NEB, E7546)	☐ P100 pipette and tips
☐ AMPure XP Beads (AXP)	☐ Qubit dsDNA HS Assay Kit (Invitrogen, Q32851)	☐ P10 pipette and tips
	☐ Freshly prepared 80% ethanol in nuclease-free water	☐ Thermal cycler
	☐ Nuclease-free water (e.g. ThermoFisher, AM9937)	☐ Microfuge
	☐ 0.2 ml thin-walled PCR tubes	☐ Hula mixer (gentle rotator mixer)
	☐ 1.5 ml Eppendorf DNA LoBind tubes	☐ Magnetic rack
	☐ Qubit™ Assay Tubes (Invitrogen, Q32856)	☐ Ice bucket with ice
		☐ Qubit fluorometer (or equivalent for QC check)

End-prep	Notes / Observations

CHECKPOINT

Check your flow cell.

We recommend performing a flow cell check before starting your library prep to ensure you have a flow cell with enough pores for a good sequencing run.

See the flow cell check instructions in the MinKNOW protocol for more information.

1. **Thaw DNA Control Sample (DCS) at room temperature, spin down, mix by pipetting, and place on ice.**

2. **Prepare the NEBNext Ultra II End Repair / dA-tailing Module reagents in accordance with manufacturer's instructions, and place on ice:**

For optimal performance, NEB recommend the following:

1. Thaw all reagents on ice.
2. Ensure the reagents are well mixed.
   Note: Do not vortex the Ultra II End Prep Enzyme Mix.
3. Always spin down tubes before opening for the first time each day.
4. The NEBNext Ultra II End Prep Reaction Buffer may contain a white precipitate. If this occurs, allow the mixture(s) to come to room temperature and pipette the buffer several times to break up the precipitate, followed by a quick vortex to mix.

3. **Prepare the DNA in nuclease-free water:**

- Transfer 100-200 fmol of amplicon DNA into a 1.5 ml Eppendorf DNA LoBind tube
- Adjust the volume to 49 µl with nuclease-free water
- Mix thoroughly by pipetting up and down, or by flicking the tube
- Spin down briefly in a microfuge

4. **In a 0.2 ml thin-walled PCR tube, mix the following:**

Between each addition, pipette mix 10-20 times.

Reagent	Volume
DNA CS	1 µl
DNA	49 µl
Ultra II End-prep Reaction Buffer	7 µl
Ultra II End-prep Enzyme Mix	3 µl

Reagent	Volume
Total	**60 µl**

5. Thoroughly mix the reaction by gently pipetting and briefly spinning down.
6. Using a thermal cycler, incubate at 20°C for 5 minutes and 65°C for 5 minutes.
7. Resuspend the AMPure XP Beads (AXP) by vortexing.
8. Transfer the DNA sample to a clean 1.5 ml Eppendorf DNA LoBind tube.
9. Add 60 µl of resuspended the AMPure XP Beads (AXP) to the end-prep reaction and mix by flicking the tube.
10. Incubate on a Hula mixer (rotator mixer) for 5 minutes at room temperature.
11. Prepare 500 µl of fresh 80% ethanol in nuclease-free water.
12. Spin down the sample and pellet on a magnet until supernatant is clear and colourless. Keep the tube on the magnet, and pipette off the supernatant.
13. Keep the tube on the magnet and wash the beads with 200 µl of freshly prepared 80% ethanol without disturbing the pellet. Remove the ethanol using a pipette and discard.
14. Repeat the previous step.
15. Spin down and place the tube back on the magnet. Pipette off any residual ethanol. Allow to dry for ~30 seconds, but do not dry the pellet to the point of cracking.
16. Remove the tube from the magnetic rack and resuspend the pellet in 61 µl nuclease-free water. Incubate for 2 minutes at room temperature.
17. Pellet the beads on a magnet until the eluate is clear and colourless, for at least 1 minute.
18. Remove and retain 61 µl of eluate into a clean 1.5 ml Eppendorf DNA LoBind tube.

CHECKPOINT

Quantify 1 µl of eluted sample using a Qubit fluorometer.

END OF STEP

Take forward the repaired and end-prepped DNA into the adapter ligation step. However, at this point it is also possible to store the sample at 4°C overnight.

Checklist: Adapter ligation and clean-up

Materials	Consumables	Equipment
☐ Ligation Adapter (LA)	☐ Salt-T4® DNA Ligase (NEB, M0467)	☐ Magnetic rack
☐ Ligation Buffer (LNB)	☐ NEBNext Quick Ligation Module (NEB, E6056)	☐ Microfuge
☐ Long Fragment Buffer (LFB)	☐ 1.5 ml Eppendorf DNA LoBind tubes	☐ Vortex mixer
☐ Short Fragment Buffer (SFB)	☐ Qubit™ Assay Tubes (Invitrogen, Q32856)	☐ P1000 pipette and tips
☐ AMPure XP Beads (AXP)		☐ P100 pipette and tips
☐ Elution Buffer (EB)		☐ P20 pipette and tips
		☐ P10 pipette and tips
		☐ Qubit fluorometer (or equivalent for QC check)

Adapter ligation and clean-up

Notes / Observations

TIP

We recommend using the Salt-T4® DNA Ligase (NEB, M0467).

Salt-T4® DNA Ligase (NEB, M0467) can be bought separately or is provided in the NEBNext® Companion Module v2 for Oxford Nanopore Technologies® Ligation Sequencing (catalogue number E7672S or E7672L).

The Quick T4 DNA Ligase (NEB, E6057) available in the previous version NEBNext® Companion Module for Oxford Nanopore Technologies® Ligation Sequencing (NEB, E7180S or E7180L) is also compatible, but the new recommended reagent offers more efficient and ligation.

IMPORTANT

Although third-party ligase products may be supplied with their own buffer, the ligation efficiency of the Ligation Adapter (LA) is higher when using the Ligation Buffer (LNB) supplied in the Ligation Sequencing Kit.

1. Spin down the Ligation Adapter (LA) and Salt-T4® DNA Ligase, and place on ice.
2. Thaw Ligation Buffer (LNB) at room temperature, spin down and mix by pipetting. Due to viscosity, vortexing this buffer is ineffective. Place on ice immediately after thawing and mixing.
3. Thaw the Elution Buffer (EB) at room temperature and mix by vortexing. Then spin down and place on ice.

IMPORTANT

Depending on the wash buffer (LFB or SFB) used, the clean-up step after adapter ligation is designed to either enrich for DNA fragments of >3 kb, or purify all fragments equally.

- To enrich for DNA fragments of 3 kb or longer, use Long Fragment Buffer (LFB)
- To retain DNA fragments of all sizes, use Short Fragment Buffer (SFB)

4. Thaw either Long Fragment Buffer (LFB) or Short Fragment Buffer (SFB) at room temperature and mix by vortexing. Then spin down and keep at room temperature.

5. In a 1.5 ml Eppendorf DNA LoBind tube, mix in the following order:

Between each addition, pipette mix 10-20 times.

Reagent	Volume
DNA sample from the previous step	60 µl
Ligation Adapter (LA)	5 µl
Ligation Buffer (LNB)	25 µl
Salt-T4® DNA Ligase	10 µl
Total	**100 µl**

6. Thoroughly mix the reaction by gently pipetting and briefly spinning down.

7. Incubate the reaction for 10 minutes at room temperature.

8. Resuspend the AMPure XP Beads (AXP) by vortexing.

9. Add 40 µl of resuspended AMPure XP Beads (AXP) to the reaction and mix by flicking the tube.

10. Incubate on a Hula mixer (rotator mixer) for 5 minutes at room temperature.

11. Spin down the sample and pellet on a magnet. Keep the tube on the magnet, and pipette off the supernatant when clear and colourless.

12. Wash the beads by adding either 250 µl Long Fragment Buffer (LFB) or 250 µl Short Fragment Buffer (SFB). Flick the beads to resuspend, spin down, then return the tube to the magnetic rack and allow the beads to pellet. Remove the supernatant using a pipette and discard.

13. Repeat the previous step.

14 Spin down and place the tube back on the magnet. Pipette off any residual supernatant. Allow to dry for ~30 seconds, but do not dry the pellet to the point of cracking.

15 Remove the tube from the magnetic rack and resuspend the pellet in 25 µl Elution Buffer (EB). Spin down and incubate for 10 minutes at room temperature. For high molecular weight DNA, incubating at 37°C can improve the recovery of long fragments.

16 Pellet the beads on a magnet until the eluate is clear and colourless, for at least 1 minute.

17 Remove and retain 25 µl of eluate containing the DNA library into a clean 1.5 ml Eppendorf DNA LoBind tube.

Dispose of the pelleted beads

CHECKPOINT

Quantify 1 µl of eluted sample using a Qubit fluorometer.

18 Depending on your DNA library fragment size, prepare your final library in 32 µl of Elution Buffer (EB).

Fragment library length	Flow cell loading amount
Very short (<1 kb)	100 fmol
Short (1-10 kb)	35–50 fmol
Long (>10 kb)	300 ng

Note: If the library yields are below the input recommendations, load the entire library.

If required, we recommend using a mass to mol calculator such as the NEB calculator.

END OF STEP

The prepared library is used for loading into the flow cell. Store the library on ice or at 4°C until ready to load.

Checklist: Priming and loading the PromethION Flow Cell

Materials	Consumables	Equipment
☐ Sequencing Buffer (SB)	☐ PromethION Flow Cell	☐ PromethION device
☐ Library Beads (LIB)	☐ 1.5 ml Eppendorf DNA LoBind tubes	☐ PromethION Flow Cell Light Shield
☐ Library Solution (LIS)		☐ P1000 pipette and tips
☐ Flow Cell Tether (FCT)		☐ P200 pipette and tips
☐ Flow Cell Flush (FCF)		☐ P20 pipette and tips

Priming and loading the PromethION Flow Cell

Notes / Observations

IMPORTANT

This kit is only compatible with R10.4.1 flow cells (FLO-PRO114M).

1. Thaw the Sequencing Buffer (SB), Library Beads (LIB) or Library Solution (LIS, if using), Flow Cell Tether (FCT) and Flow Cell Flush (FCF) at room temperature before mixing by vortexing. Then spin down and store on ice.
2. To prepare the flow cell priming mix, combine Flow Cell Tether (FCT) and Flow Cell Flush (FCF), as directed below. Mix by vortexing at room temperature.

Note: We are in the process of reformatting our kits with single-use tubes into a bottle format. Please follow the instructions for your kit format.

Single-use tubes format: Add 30 µl Flow Cell Tether (FCT) directly to a tube of Flow Cell Flush (FCF).

Bottle format: In a clean suitable tube for the number of flow cells, combine the following reagents:

Reagent	Volume per flow cell
Flow Cell Flush (FCF)	1,170 µl
Flow Cell Tether (FCT)	30 µl
Total volume	**1,200 µl**

IMPORTANT
After taking flow cells out of the fridge, wait 20 minutes before inserting the flow cell into the PromethION for the flow cell to come to room temperature. Condensation can form on the flow cell in humid environments. Inspect the gold connector pins on the top and underside of the flow cell for condensation and wipe off with a lint-free wipe if any is observed. Ensure the heat pad (black pad) is present on the underside of the flow cell.

3 For PromethION 2 Solo, load the flow cell(s) as follows:

1. Place the flow cell flat on the metal plate.

2. Slide the flow cell into the docking port until the gold pins or green board cannot be seen.

4 For the PromethION 24/48, load the flow cell(s) into the docking ports:

1. Line up the flow cell with the connector horizontally and vertically before smoothly inserting into position.
2. Press down firmly onto the flow cell and ensure the latch engages and clicks into place.

IMPORTANT
Insertion of the flow cells at the wrong angle can cause damage to the pins on the PromethION and affect your sequencing results. If you find the pins on a PromethION position are damaged, please contact support@nanoporetech.com for assistance.

5 Slide the inlet port cover clockwise to open.

IMPORTANT
Take care when drawing back buffer from the flow cell. Do not remove more than 20-30 µl, and make sure that the array of pores are covered by buffer at all times. Introducing air bubbles into the array can irreversibly damage pores.

6 After opening the inlet port, draw back a small volume to remove any air bubbles:

1. Set a P1000 pipette tip to 200 µl.
2. Insert the tip into the inlet port.
3. Turn the wheel until the dial shows 220-230 µl, or until you see a small volume of buffer entering the pipette tip.

7. Load 500 µl of the priming mix into the flow cell via the inlet port, avoiding the introduction of air bubbles. Wait five minutes. During this time, prepare the library for loading using the next steps in the protocol.

8. Thoroughly mix the contents of the Library Beads (LIB) by pipetting.

> **IMPORTANT**
> **The Library Beads (LIB) tube contains a suspension of beads. These beads settle very quickly. It is vital that they are mixed immediately before use.**
>
> We recommend using the Library Beads (LIB) for most sequencing experiments. However, the Library Solution (LIS) is available for more viscous libraries.

9. In a new 1.5 ml Eppendorf DNA LoBind tube, prepare the library for loading as follows:

Reagent	Volume per flow cell
Sequencing Buffer (SB)	100 µl
Library Beads (LIB) thoroughly mixed before use, or Library Solution (LIS)	68 µl
DNA library	32 µl
Total	**200 µl**

Note: Library loading volume has been increased to improve array coverage.

10. Complete the flow cell priming by slowly loading 500 µl of the priming mix into the inlet port.

11. Mix the prepared library gently by pipetting up and down just prior to loading.

12. Load 200 µl of library into the inlet port using a P1000 pipette.

13. Close the valve to seal the inlet port.

IMPORTANT

Install the light shield on your flow cell as soon as library has been loaded for optimal sequencing output.

We recommend leaving the light shield on the flow cell when library is loaded, including during any washing and reloading steps. The shield can be removed when the library has been removed from the flow cell.

14 If the light shield has been removed from the flow cell, install the light shield as follows:

1. Align the inlet port cut out of the light shield with the inlet port cover on the flow cell. The leading edge of the light shield should sit above the flow cell ID.
2. Firmly press the light shield around the inlet port cover. The inlet port clip will click into place underneath the inlet port cover.

END OF STEP

Close the PromethION lid when ready to start a sequencing run on MinKNOW.

Wait a minimum of 10 minutes after loading the flow cells onto the PromethION before initiating any experiments. This will help to increase the sequencing output.

Checklist: Flow cell reuse and returns

Materials	Consumables	Equipment
☐ Flow Cell Wash Kit (EXP-WSH004)		

Flow cell reuse and returns	Notes / Observations

1. After your sequencing experiment is complete, if you would like to reuse the flow cell, please follow the Flow Cell Wash Kit protocol and store the washed flow cell at +2°C to +8°C.

The Flow Cell Wash Kit protocol is available on the Nanopore Community.

> **TIP**
> **We recommend you to wash the flow cell as soon as possible after you stop the run. However, if this is not possible, leave the flow cell on the device and wash it the next day.**

2. Alternatively, follow the returns procedure to send the flow cell back to Oxford Nanopore.

Instructions for returning flow cells can be found here.

> **IMPORTANT**
> **If you encounter issues or have questions about your sequencing experiment, please refer to the Troubleshooting Guide that can be found in the online version of this protocol.**